

Hamza Arfin

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ABOUT

Full-stack developer experienced in building web applications using Python, Flask, and MySQL. Skilled in REST API development, database design, and backend systems with projects in recommendation systems, NLP, and data-driven applications.

EXPERIENCE

FLAme NLP Research Lab, IIIT Delhi

July 2025 – September 2025

Undergraduate Research Intern

New Delhi, India

- Developed and deployed the official lab website (FLAme NLP Lab) using **Jekyll**
- Built a scalable static website with modular components for managing research updates, publications, and team information
- Structured content using templates and data files to enable easy updates without modifying core code
- Improved usability, performance, and maintainability of the lab's digital presence

Software Development Projects

2023 – Present

Full-Stack Developer

New Delhi, India

- Built full-stack web applications using **Python, Flask, MySQL, HTML, CSS, JavaScript**
- Developed **REST APIs**, authentication systems, and database workflows
- Designed backend architecture and relational schemas using **MySQL**
- Implemented responsive UI and managed version control with **Git/GitHub**

PROJECTS

BERT Sarcasm Detection Classifier | GitHub

May 2026

- Fine-tuned **bert-base-uncased** for sarcasm detection using headline and balanced sarcasm datasets
- Trained **BertForSequenceClassification** with class weighting and optimized threshold tuning
- Reached **93.77% F1, 94.06% accuracy, 0.983 ROC-AUC, and 0.979 PR-AUC**
- Generated confusion matrix, ROC/PR curves, training curves, and final model artifacts

Hybrid Movie Recommendation System | GitHub

March 2026

- Built a recommendation platform using **Python, Flask, HTML, CSS, JavaScript**
- Implemented hybrid recommender using **KNN collaborative filtering** on the **MovieLens dataset (100k+ ratings)**
- Integrated **TMDB API** and built REST endpoints serving movie metadata and posters

Streamly – Streaming Aggregator | GitHub

November 2025

- Developed a streaming search platform using **Flask** and the **TMDB API**
- Implemented real-time search suggestions and optimized API requests with rate limiting

Flask Hospital Management System

August 2024

- Built hospital management web app using **Flask, MySQL, SQLAlchemy**
- Designed relational database schema managing patient, doctor, and appointment records across multiple tables

Python–MySQL Hospital Database System

October 2023

- Developed hospital database system using **Python** and **MySQL**
- Implemented CRUD operations for patient, doctor, and appointment management

EDUCATION

Jamia Hamdard University

2023 – 2027

Bachelor of Technology in Computer Science and Engineering

New Delhi, India

- **Relevant Coursework:** Data Structures, Algorithms, Database Systems, Software Engineering
- Focused on web development, backend systems, and database-driven applications

Dev Samaj Modern School - 2

2021 – 2023

Senior Secondary School

New Delhi, India

Dev Samaj Modern School - 2

2019 – 2021

Secondary School

New Delhi, India

SKILLS

Languages: Python, C, C++, JavaScript, SQL

Web: Flask, HTML, CSS, REST APIs, Backend Development, Jekyll

ML/NLP: NumPy, Pandas, Scikit-learn, BERT, Transformers, PyTorch

Database: MySQL, SQLAlchemy

Tools: Git, GitHub

Concepts: Data Structures, Algorithms, CRUD, API Integration